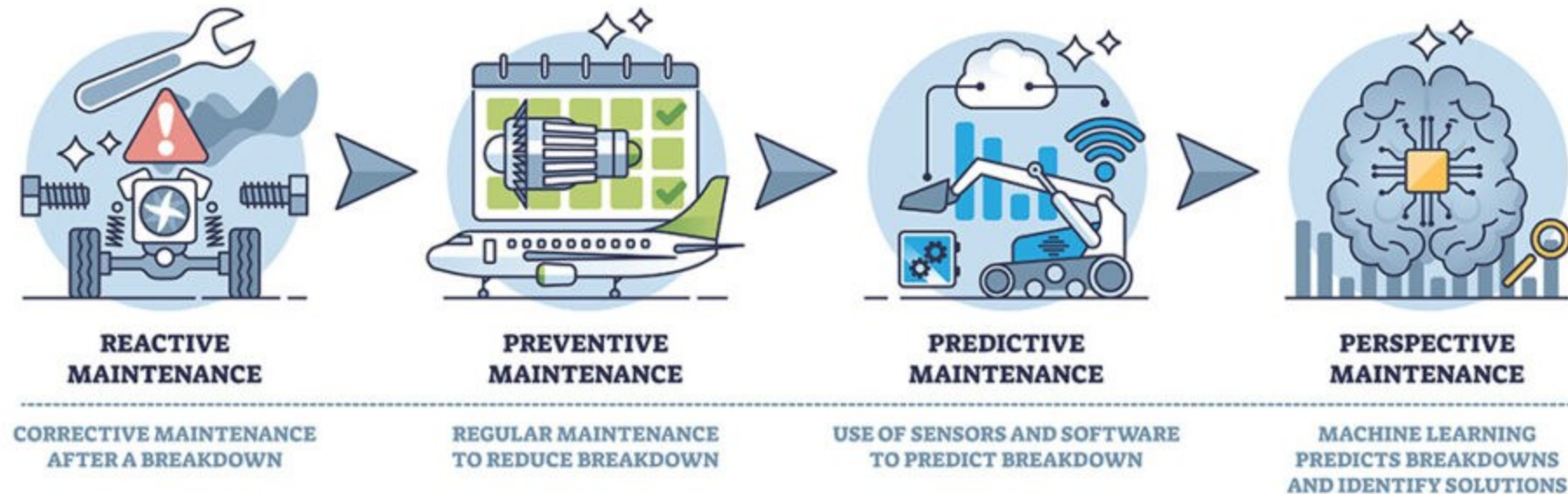


AIoT Machine Learning - Predictive Maintenance

PREDICTIVE MAINTENANCE



- Predict failures before they occur
- Reduce downtime & operational losses
- Enable intelligent asset management

AIoT Machine Learning - Predictive Maintenance

Maintenance Approaches



Reactive → fix after failure



Preventive → scheduled maintenance



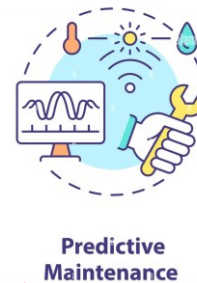
Predictive → data-driven maintenance

Why Predictive Maintenance Matters



Unplanned Downtime is Costly

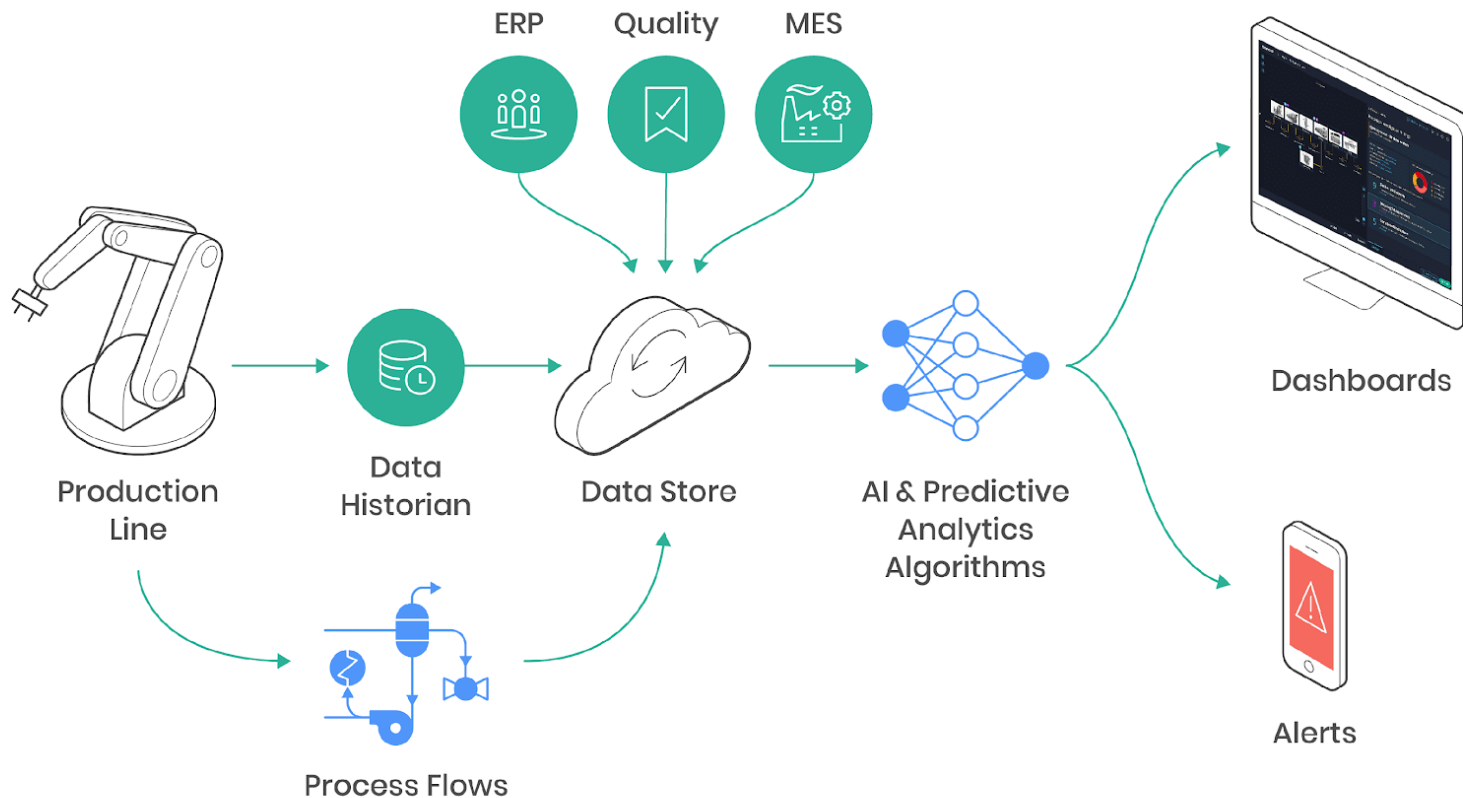
- Production loss
- Service disruption
- Safety risks



Predictive Maintenance Helps

- ✓ Reduce downtime
- ✓ Extend equipment life
- ✓ Improve safety
- ✓ Optimize maintenance schedules

How Predictive Maintenance Works



Sensors



Data Collection



AI Analysis



Failure Prediction



Maintenance Action

Data Used in Predictive Maintenance

Common Sensor Inputs



Vibration signals



Temperature readings



Pressure levels



Electrical current & voltage

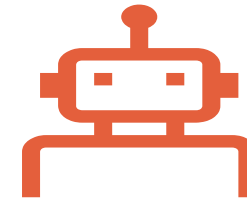


Acoustic signals

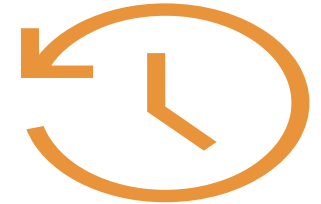
Operational Data



Usage
patterns

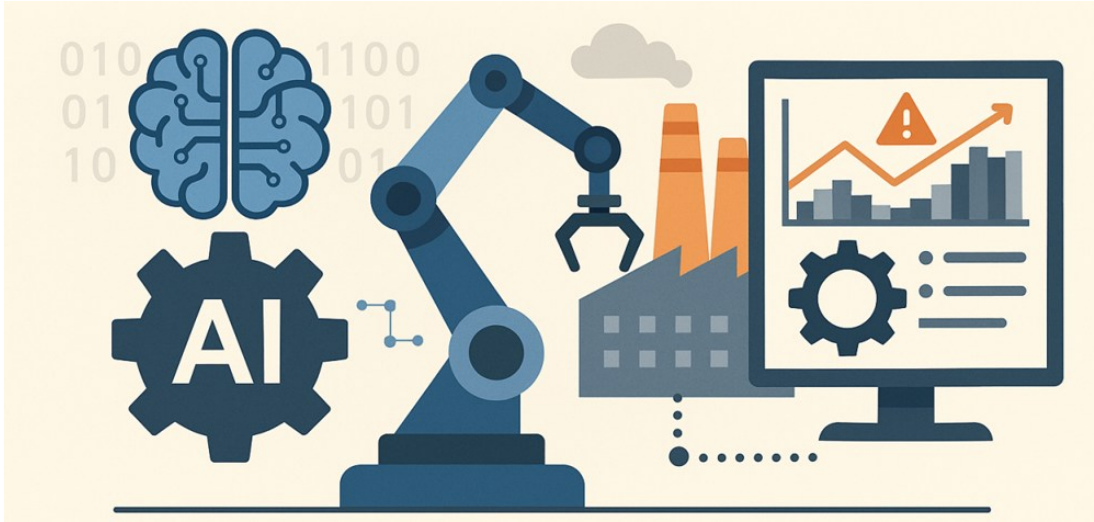


Load
conditions



Maintenance
history

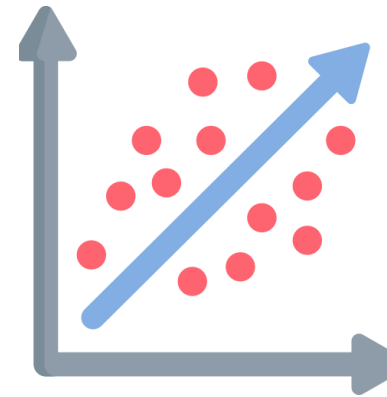
Machine Learning Models Used



AI Models Learn Patterns Like:

- ⚠ Increasing vibration amplitude
- ⚠ Rising temperature trends
- ⚠ Energy consumption spikes
- ⚠ Irregular operating cycles
- ⚠ Performance degradation over time

Machine Learning Models Used



Regression Models

- Predict remaining useful life (RUL)
 - ✓ Linear Regression
 - ✓ Random Forest Regression
 - ✓ Gradient Boosting

Classification Models

- Predict failure likelihood
 - ✓ Logistic Regression
 - ✓ Decision Trees
 - ✓ Neural Networks



Machine Learning Models Used

Real-World Use Cases



Energy → wind turbine health monitoring



Railways → wheel & track wear prediction



Oil & Gas → pipeline failure prevention



Aviation → engine health monitoring

Benefits to Organizations



Operational Benefits

- ✓ Reduced downtime
- ✓ Lower maintenance costs
- ✓ Improved equipment lifespan
- ✓ Better spare parts planning

Strategic Benefits

- ✓ Increased productivity
- ✓ Improved reliability
- ✓ Enhanced safety compliance



Implementation Challenges & Next Step

Common Implementation Challenges

- ⚠ Poor data quality
- ⚠ Insufficient failure history
- ⚠ Sensor calibration issues
- ⚠ Model drift over time
- ⚠ Integration with legacy systems

Evaluating ML Models in AIoT

**Building models is not enough.
We must evaluate**

- Accuracy
- Reliability
- False alarm rates
- Real-world performance